TrachitZ YouTube Channel Analysis: Remote LLM Setup Guide

Introduction

This document provides a comprehensive analysis of the TrachitZ YouTube channel, focusing on content related to Large Language Model (LLM) setup, particularly on cloud infrastructure. The primary goal is to extract relevant information to guide users in establishing a powerful remote LLM setup accessible from any computer, addressing the common challenge of local hardware limitations.

1. All Videos on the Channel

The TrachitZ YouTube channel features 40 videos covering a range of technical topics. Below is a list of all videos with their titles and brief descriptions, extracted from the channel page.

Video Title	Views	Upload Date	Description (Summary)
I built a FREE AI API That DESTROYS OpenAI and other AI with 100K Requests/Day	3.1K	5 months ago	Tutorial on building a free text generation API using Cloudflare Workers AI, offering 100,000 requests/day. Covers setup, configuration, and use of various LLMs.
The #1 FREE AI Image Generation API You Need to Know About	6.1K	6 months ago	Tutorial on setting up a free AI image generation API using Cloudflare Workers, capable of generating 100,000 images/day.
ESP32 / ESP8266 Advanced Web Server with File Upload and Hosting!	1.1K	11 months ago	Guide to creating an advanced web server using ESP32/ESP8266 with file upload and hosting capabilities.
Make Simple Web Server using ESP32 / ESP8266 board	979	11 months ago	Basic tutorial on setting up a simple web server with ESP32/ESP8266.
Arduino IDE Setup for ESP8266 & ESP32: Complete Step-by-Step Tutorial (2025)	822	11 months ago	Step-by-step guide to setting up Arduino IDE for ESP8266 and ESP32 boards.
GET FREE SSL CERTIFICATE WITH CERTBOT NOW!	Let’s Encrypt SSL	FeroHost	606
Sneak Peek at Snaps WITHOUT Being Caught!	442	1 year ago	(Description not available from initial scrape)
WIFI Security Testing - Exploring OneShot	Check and Fix WPS Vulnerability	2.8K	1 year ago
I Started My Own Free Web Hosting Service	FeroHost	Best Free Web Hosting in 2025	1.8K

Video Title	Views	Upload Date	Description (Summary)
SQLite Database for Java in IntelliJ IDEA	Step-by-Step Tutorial with CRUD Operations	8.8K	1 year ago
IntelliJ IDEA for Java: Install and Set Up	Free Student License Tutorial	25K	1 year ago
Portfolio Website: No code, Free	WordPress Customization	466	1 year ago
Minecraft Server for Free 🚀	Not Aternos	6GB RAM Server Free	Step-by-Step Guide
Dynamic and Personalized Chrome with Cool Themes! 🌐💫	Using Chrome Extention	TrachitZ	183
Left Right Speaker Test	Check your Audio Device	Tobu-Candyland	TrachitZ
Exploring zPhisher and Protecting Yourself from Phishing Attacks	TrachitZ	289	2 years ago
Installing WordPress in XAMPP and Setting Up XAMPP with Database	Complete Setup TrachitZ	237	2 years ago
Dual Boot Kali Linux with Windows 11	In new Laptop	Install Kali Linux Full Process	TrachitZ
Windows 11 Installation	In New Acer Aspire 5 Laptop	Create Bootable USB	TrachitZ
Windows 10 in Ten years old Laptop 😲 💀	SSD Installation	Windows 10 Installation	TrachitZ

Video Title	Views	Upload Date	Description (Summary)
Type In NEPALI (English Text)	Easiest Way	Google Input Tools Install and use	TrachitZ
5 Years Old CPU FIXED	CMOS Battery Replacement	43	2 years ago
Make Bootable USB	Kali Linux 💀 💀 Live USB	TrachitZ	69
Turn OLD Laptop into POWERFUL Ubuntu Server in 7 Minutes	746	2 years ago	Tutorial on converting an old laptop into an Ubuntu server for various uses.

2. Most Relevant Videos for Setting Up LLMs on AWS or Cloud Infrastructure

Based on the user's request and the channel's content, the most relevant videos are those discussing AI APIs and cloud-based setups. It is important to note that **no videos explicitly mention AWS services** for LLM deployment. Instead, the channel focuses on **Cloudflare Workers AI** as a free and powerful alternative for running LLMs remotely.

The key videos identified are:

- **"I built a FREE AI API That DESTROYS OpenAI and other AI with 100K Requests/Day"** [1]
    - **Relevance:** Highly relevant. This video directly addresses the user's need for a powerful, remote LLM setup without local hardware constraints. It details how to leverage Cloudflare Workers AI for free text generation with a high request limit.
    - **Key Topics:** Free text generation API, Cloudflare Workers AI, 100,000 free daily requests, API key authentication, serverless API, chatbot development.

- **LLM Models Mentioned:** OpenAI 120B, Meta Llama 70B, Mistral 7B, DeepSeek, Gemma, Llama 4 Scout, DeepSeek R1, Mistral Small 3.1, Qwen, Microsoft Phi-2.
- **Tools/Frameworks:** Cloudflare Workers, Cloudflare Dashboard, JavaScript, GitHub, cURL.

- **"Run ANY LLM Without GPU for Free on Cloud (Llama, Gemini, Claude & More!) #shorts #ai"** [2]
  - **Relevance:** Highly relevant. This short video acts as a teaser for the full tutorial mentioned above, emphasizing the ability to run various LLMs on the cloud without a GPU.
  - **Key Topics:** Running LLMs on cloud, GPU-free operation, free API key, open-source LLMs.
  - **LLM Models Mentioned:** GPT-OSS, Llama 3, DeepSeek-R1, Gemini, Claude, MistralAI, Google Gemma, Microsoft Phi-2, Qwen.
  - **Tools/Frameworks:** Cloudflare (implied).

- **"The #1 FREE AI Image Generation API You Need to Know About"** [3]
  - **Relevance:** Moderately relevant. While focused on image generation, this video demonstrates the same underlying Cloudflare Workers AI technology for deploying AI models, which is directly applicable to LLMs. It reinforces the concept of free, cloud-based AI APIs.
  - **Key Topics:** AI image generation, Cloudflare Workers, API setup, API key authentication, serverless functions.
  - **LLM Models Mentioned:** `@cf/stabilityai/stable-diffusion-xl-base-1.0`, `@cf/blackforestlabs/ux-l-schnell`, `@cf/bytedance/stable-diffusion-xl-lightning`, `@cf/lykon/dreamshaper-8-lcm`, `@cf/runwayml/stable-diffusion-v1-5-img2img`, `@cf/runwayml/stable-diffusion-v1-5-inpainting`.
  - **Tools/Frameworks:** Cloudflare Workers, Workers AI, cURL, JavaScript.

- **"Turn OLD Laptop into POWERFUL Ubuntu Server in 7 Minutes"** [4]
  - **Relevance:** Low relevance to LLMs, but relevant to the concept of remote computing. This video shows how to repurpose an old laptop into a server,

which could theoretically host smaller LLMs or act as a gateway, but it does not cover LLM deployment.
    - **Key Topics:** Ubuntu Server, repurposing old hardware, remote access (OpenSSH), web hosting.
    - **Tools/Frameworks:** Ubuntu Server 22.04.1 LTS, OpenSSH, WordPress, Nextcloud.

3. Key Methods, Tools, and Services Mentioned

The primary method advocated by TrachitZ for running powerful AI models remotely is **Cloudflare Workers AI**. This platform allows users to deploy serverless functions (Workers) that can interact with a wide range of pre-trained AI models hosted by Cloudflare.

Key Tools and Services:

- **Cloudflare Workers AI:** A serverless platform that provides access to various AI models for inference at the edge. It offers a generous free tier, making it an attractive option for developers and individuals.
- **Cloudflare Dashboard:** The web interface for managing Cloudflare accounts, Workers, and AI bindings.
- **JavaScript:** The programming language used to write the Cloudflare Worker scripts.
- **GitHub:** Used for hosting the worker code and providing setup instructions.
- **cURL/Browser Developer Console:** Tools for testing the deployed API endpoints.

LLM Models Supported (via Cloudflare Workers AI): The videos highlight access to a diverse set of LLMs, including:

- **Text Generation:** OpenAI (gpt-oss-120b), Meta Llama (llama-3.3-70b-instruct-fp8-fast, Llama 4 Scout 17B), MistralAI (mistral-7b-instruct-v0.2, Mistral Small 3.1 24B), DeepSeek (deepseek-r1-distill-qwen-32b, DeepSeek R1), Google Gemma (gemma-3-12b-it), Microsoft Phi-2, Qwen (qwen2.5-coder-32b-instruct, qwen1.5-14b-chat-awq).

- **Image Generation:** Stable Diffusion XL models (e.g., `@cf/stabilityai/stable-diffusion-xl-base-1.0`).

Absence of AWS: It is crucial to note that despite the user's initial mention of AWS, the TrachitZ channel's content for free remote LLM setup exclusively focuses on Cloudflare Workers AI. There is no mention of AWS EC2, SageMaker, Bedrock, or other AWS services in the context of LLM deployment in the analyzed videos.

4. Practical Guide: Setting Up a Powerful Remote LLM with Cloudflare Workers AI

Based on the TrachitZ channel's content, here is a practical guide to setting up a powerful remote LLM that can be accessed from any computer, leveraging Cloudflare Workers AI.

Step-by-Step Setup:

1. **Create a Cloudflare Account:**
    - Sign up for a free account on the Cloudflare website.
2. **Create a New Cloudflare Worker:**
    - Navigate to the Cloudflare Workers dashboard.
    - Click on **"Create application"** and then **"Create Worker."**
    - Give your Worker a descriptive name (e.g., `my-llm-api`).
    - Click **"Deploy"** to create a basic "Hello World" Worker.
3. **Replace the Worker Code:**
    - Access the Worker editor for your newly created Worker.
    - Replace the default "Hello World" code with the script provided in the TrachitZ GitHub repository for the free text generation API.
    - Click **"Save and Deploy."**
4. **Set Up Environment Variables:**
    - In your Worker's dashboard, go to **"Settings"** > **"Variables."**

- Under **“Environment Variables,”** click **“Add variable.”**
- Set **Name:** `API_KEY`.
- Set **Value:** Choose a strong, secret key (e.g., a long, random string). This will be used to authenticate requests to your API.
- Click **“Save and Deploy.”**

5. **Enable Workers AI:**
   - In the main Cloudflare dashboard, navigate to **“Workers & Pages”** > **“AI.”**
   - Enable Workers AI for your account. The free tier is sufficient for this setup.
6. **Add AI Binding to Your Worker:**
   - Go back to your Worker’s dashboard (**“Settings”** > **“Variables”**).
   - Scroll down to the **“Service bindings”** section.
   - Click **“Add binding.”**
   - Set **Variable name:** `AI`.
   - For **Service:** Select **“Workers AI”** from the dropdown menu.
   - Click **“Save and Deploy.”**
   - **Important:** This binding is crucial for your Worker to access Cloudflare’s AI models.
7. **Get Your Worker URL:**
   - Your deployed Worker will have a unique URL, typically in the format `https://<your-worker-name>.<your-subdomain>.workers.dev`.
   - You can find this exact URL in your Worker’s dashboard.

Usage and Testing:

Once deployed, you can interact with your remote LLM API using cURL or JavaScript:

cURL Example:

```
curl -X POST https://<your-worker-name>.<your-subdomain>.workers.dev \\
  -H "Authorization: Bearer your-secret-api-key" \\
  -H "Content-Type: application/json" \\
  -d '{
    "prompt": "Write a short poem about a rainy day.",
    "systemPrompt": "You are a helpful assistant.",
    "history": [
      { "role": "user", "content": "Tell me something about weather." },
      { "role": "assistant", "content": "Weather is the state of the
atmosphere." }
    ]
  }'
```

JavaScript Example (in browser console or Node.js):

```
const res = await fetch("https://<your-worker-name>.<your-
subdomain>.workers.dev", {
  method: "POST",
  headers: {
    "Authorization": "Bearer your-secret-api-key",
    "Content-Type": "application/json",
  },
  body: JSON.stringify({
    prompt: "What is the capital of France?",
    systemPrompt: "You are a knowledgeable assistant.",
    history: [
      { role: "user", content: "Tell me about Europe." },
      { role: "assistant", content: "Europe is a continent with many
countries." }
    ]
  }),
});
const text = await res.text();
console.log(text);
```

Parameters for API Calls:

- `prompt` (string, required): The main input text for the LLM.
- `systemPrompt` (string, optional): Instructions to guide the LLM's behavior (e.g., "You are a helpful assistant.").

- `history` (array of objects, optional): An array of previous `user` and `assistant` messages to maintain conversation context.

5. Estimated Costs, Hardware Requirements, and Tools/Frameworks

Estimated Costs:

The primary advantage of this setup, as highlighted by TrachitZ, is its **cost-effectiveness**. Cloudflare Workers AI offers a **free tier** that includes:

- **100,000 AI requests per day** for text generation and image generation.
- A significant number of Worker invocations and CPU time.

This makes it an excellent solution for personal projects, learning, and even many production-ready applications without incurring direct costs. For usage beyond the free tier, Cloudflare's pricing is generally considered competitive, but the videos emphasize the free aspect for the described setup.

Hardware Requirements:

None for the user's local machine. This is the core benefit of a remote LLM setup. The LLM processing is handled entirely by Cloudflare's global network of edge servers. The user's local computer only needs to be able to send HTTP requests and receive responses, which can be done from virtually any internet-connected device.

Instance Types (Cloudflare's Infrastructure): While specific instance types are not detailed, Cloudflare Workers run on a highly distributed, serverless infrastructure. This means users do not need to provision or manage virtual machines (like AWS EC2 instances) or specialized hardware (like GPUs). Cloudflare handles all the underlying infrastructure, scaling, and model serving.

Tools and Frameworks Recommended:

- **Cloudflare Workers:** The serverless platform for deploying the API logic.
- **Cloudflare Workers AI:** The service providing access to the LLM models.

- **JavaScript:** For writing the Worker code.
- **GitHub:** For version control and accessing the provided worker script.
- **cURL:** A command-line tool for making HTTP requests and testing the API.
- **Browser Developer Console:** For interactive testing of the API using JavaScript `fetch` requests.
- **Visual Studio Code (or any code editor):** For editing the `worker.js` file.

6. Specific Models Recommended and Why

The TrachitZ channel, through its tutorials on Cloudflare Workers AI, recommends a variety of powerful LLM models, emphasizing their accessibility and performance on the platform. The recommendation stems from the fact that these models are readily available and performant within Cloudflare's free tier.

Recommended Text Generation Models:

- **Meta Llama 4 Scout 17B:** Described as the "Latest multimodal model with 16 experts," suggesting advanced capabilities for complex tasks.
- **Meta Llama 3 (70B-instruct-fp8-fast):** A powerful open-source model known for its strong performance in instruction following and general text generation.
- **MistralAI (Mistral 7B-instruct-v0.2, Mistral Small 3.1 24B):** Mistral models are highly regarded for their balance of performance and efficiency, making them excellent choices for various applications.
- **DeepSeek (DeepSeek R1 Distill 32B, DeepSeek-Math 7B-instruct):** DeepSeek models offer strong reasoning capabilities, particularly the math-focused variant, which is valuable for technical applications.
- **Google Gemma 3 12B-IT:** Google's open-source model, known for its quality and efficiency.
- **OpenAI (gpt-oss-120b):** A large, capable model for general text generation tasks.
- **Qwen (Qwen2.5-Coder-32B-instruct, Qwen1.5-14B-chat-awq):** Qwen models, especially the coder variant, are suitable for programming-related tasks and chat applications.
- **Microsoft Phi-2:** A smaller yet capable model, often used for its efficiency.

Why these models are recommended:

- **Accessibility:** They are available through Cloudflare Workers AI, often within the free tier.
- **Performance:** They offer competitive performance for various text generation tasks, rivaling commercial APIs.
- **Cost-Effectiveness:** Utilizing these models via Cloudflare's free tier eliminates the need for expensive GPU hardware or paid API subscriptions.
- **Variety:** The wide selection allows users to choose models best suited for their specific application (e.g., general chat, coding, reasoning).

Conclusion

The TrachitZ YouTube channel provides valuable insights into setting up powerful remote LLM capabilities without relying on local GPU hardware or expensive cloud services like AWS. The core strategy revolves around leveraging **Cloudflare Workers AI** to deploy serverless API endpoints that can access a diverse range of open-source and proprietary LLMs. This approach offers a highly cost-effective solution with a generous free tier, making advanced AI accessible to a broader audience. While the channel does not delve into AWS-specific LLM deployments, its focus on Cloudflare presents a compelling alternative for remote LLM infrastructure.

References

[1] I built a FREE AI API That DESTROYS OpenAI and other AI with 100K Requests/Day - YouTube [2] Run ANY LLM Without GPU for Free on Cloud (Llama, Gemini, Claude & More!) #shorts #ai - YouTube [3] The #1 FREE AI Image Generation API You Need to Know About - YouTube [4] Turn OLD Laptop into POWERFUL Ubuntu Server in 7 Minutes - YouTube [5] GitHub - saurav-z/free-text-generation-api [6] Cloudflare Workers dashboard [7] Cloudflare website